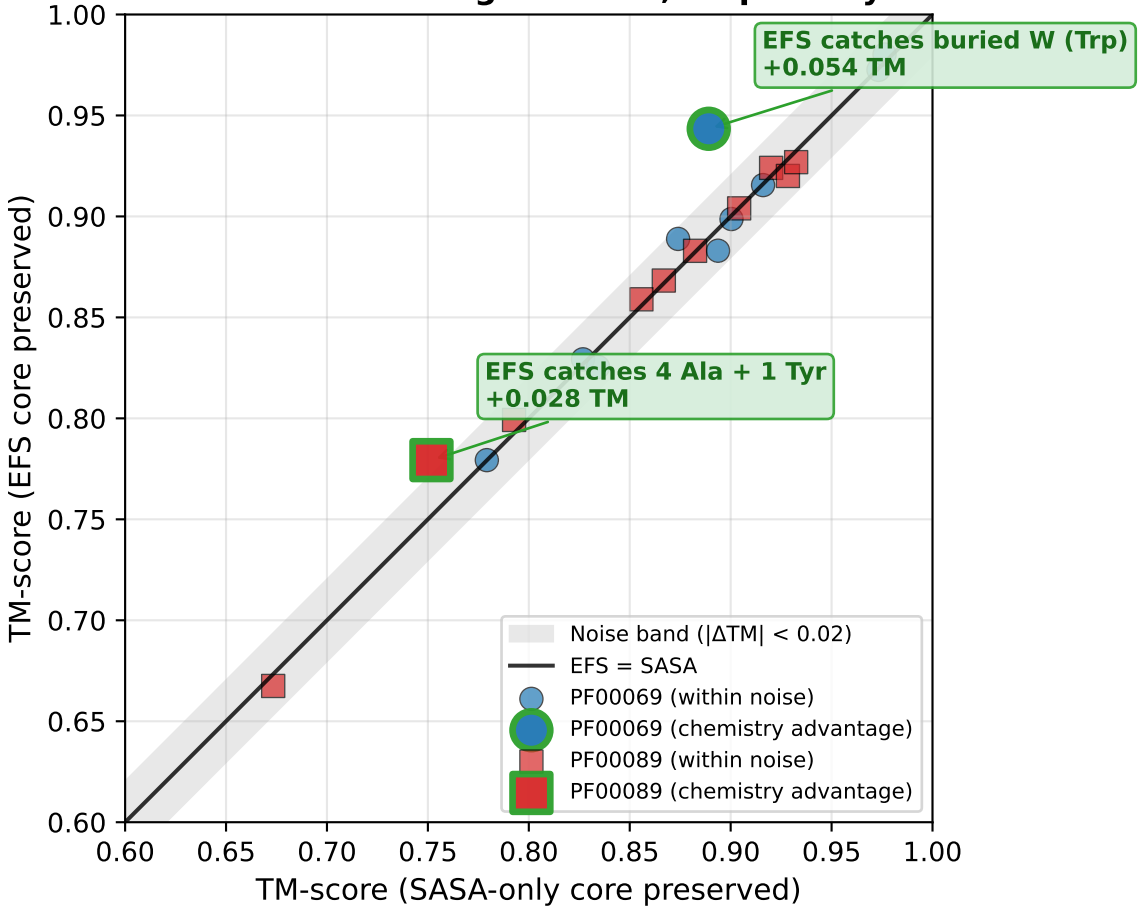
EFS subsumes SASA as a special case — paired folding test

A. EFS ≥ SASA: theoretical guarantee, empirically confirmed



Why EFS is always ≥ SASA

```
EFS raw_score = [SASA burial bonus]
              + [contact bonus]
              + [hydrophobic + buried bonus]
              + [cysteine bonus]
              + [aromatic + buried bonus]
              ──────────────────────────
              ≥ SASA criterion alone
```

EFS uses SASA AND additional chemistry signals.
By construction: EFS catches every position SASA does,
plus some that SASA misses.

Predicted outcomes by family chemistry:

- **Chemistry-rich families** → **EFS wins clearly**
  (disulfide-bridged, aromatic-stacked, tightly packed hydrophobic cores)
- **Chemistry-mild families** → **EFS reduces to SASA (tied)**
  (loose folds, primarily SASA-driven burial)
- **Either way** → **SASA never wins above noise**
  (by mathematical construction)

Empirical: 0 SASA wins above noise across 20 paired comparisons.